Mayank Bharwal

Toronto, ON | (647) 594-2129 | mayankjobapp27@duck.com | linkedin.com/in/mayank-bharwal-68802b283 | github.com/bharwalmayank | bharwalmayank.github.io

Education

University of Toronto - St. George **Expected Graduation: May 2028**

Honours Bachelor of Science in Computer Science (AI and Systems Dual Focus) *Toronto, ON*

- **Awards:** Dean's List Scholar, The Regents In-Course Scholarship
- **Relevant Coursework:** Computer Architecture, Systems Programming, Machine Learning, Computer Security, Web Programming, Advanced Data Structures and Algorithms, Multivariable Calculus, Concurrency and Parallelism, Databases Theory with SQL

Work Experience

TD Bank **May 2026 to Present**

Software Engineer Intern *Toronto, ON*

- Collaborated within a 7-engineer team to build a patent-pending autonomous AI platform for enterprise-wide application monitoring, automated testing, incident response, and operational resilience across multiple business units.
- Engineered multi-agent workflows using **Python**, **LangGraph**, **LangChain**, **FastAPI**, **Selenium**, and **Playwright** to continuously validate hundreds of production applications, projecting a **5x reduction** in manual testing effort and a **10x improvement** in MTTR.
- Architected scalable backend microservices and observability pipelines using **Spring Boot**, **Docker**, **Kubernetes**, **Linux**, **Prometheus**, and **Grafana**, supporting an initiative projected to deliver **$25M+** in annual operational savings.
- Awarded 1st Place Innovation Trophy during TD Innovation Week within two weeks of onboarding for delivering the competition's highest-impact technical prototype.

Trustworthy Machine Intelligence **Oct 2025 to Present**

Software Engineer - AI Research Club *Toronto, ON*

- Simulated 50+ autonomous **LLM agents** across small-world network topologies in **Python**, achieving **35% higher** sustained cooperation across 100-round Public Goods games.
- Architected stateful agent cognitive pipelines incorporating episodic memory, persona conditioning, and LLM policy routing, improving strategic coordination accuracy by **28%** in repeated Prisoner's Dilemma benchmarks.
- Implemented algorithmic mechanism design interventions (dynamic penalties, subsidies, and cross-community graph links), boosting fairness metrics by **22%** and reducing network polarization by **15%**.

University of Toronto Amateur Rocketry Club **Aug 2025 to Present**

Software Engineer - Avionics Team Lead *Toronto, ON*

- Architected Software Abstraction (SAL), Hardware Abstraction (HAL), and automated testing layers using **Cython**, **C++**, and **JSBSim**, achieving **95%+ simulation fidelity** and a **60% improvement** in landing precision via real-time sensor fusion.
- Engineered a low-latency RF telemetry stack and ground station GUI for real-time flight systems monitoring, packet decoding, and telemetry data logging at **95% transmission accuracy**.
- Led 8 to 10 avionics developers in an Agile workflow and built **GitHub Actions CI/CD** pipelines integrated with hardware-in-the-loop (HIL) testing to automate flight software verification.

Hyundai Motor Company **Jun 2023 to Aug 2023**

Software Engineer Intern *India*

- Engineered and deployed an internal employee management web platform on on-premise servers using **React.js**, **Django**, **JavaScript**, and **HTML/CSS** to centralize workforce operations.
- Designed a relational **SQL** database schema and RESTful API endpoints to track longitudinal employee performance reviews, project allocations, and department records.
- Automated onboarding and evaluation workflows across **200+ employees** via interactive React form validation, eliminating manual spreadsheet entry and accelerating HR data collection.

Technical Skills

Languages: Python, Java, C++, C, Cython, TypeScript, JavaScript, SQL, Rust, Bash, HTML/CSS
AI & Machine Learning: LangGraph, LangChain, PyTorch, TensorFlow, scikit-learn, OpenAI API, Multi-Agent Systems
Frameworks & Libraries: Spring Boot, FastAPI, Django, React.js, Node.js, Selenium, Playwright, REST APIs, GraphQL
Cloud, DevOps & Databases: Docker, Kubernetes, AWS, Google Cloud (GCP), Linux, GitHub Actions (CI/CD), Jenkins, Prometheus, Grafana, PostgreSQL, MongoDB, SQLite, Git

Projects

YapYap - Distributed Social Messenger | *Java, MongoDB, Clean Architecture, Git* **github.com/bharwalmayank/YapYap**

- Engineered a real-time desktop messaging application in a 6-person team using **Java** and **MongoDB**, strictly adhering to **Clean Architecture** and **SOLID** principles to decouple business logic from UI and persistence layers.
- Implemented **Factory**, **Facade**, **Observer**, and **Dependency Injection** design patterns with comprehensive unit test coverage to support scalable chat rooms, user authentication, and asynchronous message updates.